Sasikaran Piraveen

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sasikaran-piraveen | piraveen/me | Piraveen



Summary

Computer Science undergraduate and Software Engineering Intern currently building AI-driven ERP automation systems, combining full-stack development experience with hands-on exposure to LLM agent workflows, backend services, and database engineering. Skilled at translating business processes into scalable, automated software solutions. Adaptable, proactive, and driven to continuously learn and innovate.

Education

Bachelor of Science Honours in Computer Science (Reading), University of Jaffna — 2022 – Present

- **Current OGPA:** 3.85/4.0 (for 96 credits out of 120 credits)
- **Dean's List:** Level 1S (2021), Level 2S (2022)

Physical Science, J/ Hartley College — 2011 – 2020

- **G.C.E. O/L:** 7A, 2B (2017)
- **G.C.E. A/L:** A, B, C; Z-score 1.3841 (2020)

Experience

Software Engineering Intern, David Pieris Holdings (Pvt) Ltd — May 2026 – Present

- Working full-time as part of a team building a company-wide AI ecosystem for ERP automation, contributing across backend development, observability for agent pipelines, and prompt engineering for LLM-driven decision-making.
- **Technologies:** Python | FastAPI | LangGraph | Azure | PostgreSQL | Oracle | Docker | Redis

Projects

Lightweight CNN-Based Deepfake Face Image Detection Using Multi-Stack Error Level Analysis — Present

- Developing a lightweight CNN model that uses multi-stack Error Level Analysis (ELA) to detect facial image manipulation through compression artifacts.
- **Technologies:** Python | PyTorch | Google Colab

Car Rental Management - AI-Enabled Booking Platform Live Demo AI Assistant Demo — Present

- Engineered a car rental platform with multi-role access control using PostgreSQL RLS and atomic conflict detection to prevent booking race conditions under concurrent reservations, built with Next.js.
- Built a natural-language fleet query assistant using an LLM query-planner and SQL AST validator for secure, read-only fleet analytics.
- **Technologies:** Next.js | TypeScript | PostgreSQL | Supabase | LLM Tool-Calling | Zustand | shadcn/ui

WhatsApp Order Management System - LLM Agent Automation WhatsApp Agent Demo — Present

- Built a WhatsApp ordering system where LLM agents propose order actions as structured tool-calls, executed only after confirmation by deterministic backend code, with prices and totals always sourced from verified database reads, never LLM-generated.
- **Technologies:** Python | FastAPI | OpenRouter | LLM Agents | PostgreSQL | WhatsApp Business API

Olirsuite - Company Management System (University Group Project) Live Demo — 2025

- Built responsive frontend modules while integrating backend APIs to implement income/expense transaction tracking and support debugging and release testing for stability.
- **Technologies:** Next.js | TypeScript | shadcn/ui | MongoDB

CheatShield – Exam Environment Control System (University Group Project) Live Demo — 2026

- Implemented allowlist-based web access control with polling-based policy synchronization, symmetric-key client registration (TTL), and secure admin-to-client policy enforcement.
- **Technologies:** Next.js | TypeScript | MongoDB | Python | Postman

Skills

Programming: Java, Python, C++, PHP, JavaScript, TypeScript
AI & Backend: LangGraph, FastAPI, PyTorch, LLM Agent Workflows, Prompt Engineering
Web Development: HTML, CSS, Tailwind CSS, React.js
Frameworks: Next.js, Nest.js
Databases: PostgreSQL, SQLite, Oracle, MySQL, MongoDB
DevOps & Tools: Docker, Redis, Microsoft Entra ID, Bitbucket, Jira, Git/GitHub, Postman
IDEs: Visual Studio, IntelliJ IDEA, MATLAB, Google Colab
Editing & Design: Canva, Capcut

Certifications

Computer Applications Assistant - NVQ Level 3 2022
Diploma in English - British College 2022
Python for Beginners - University of Moratuwa 2023
Web Design for Beginners - University of Moratuwa 2024
Introduction to Web Development with HTML5, CSS3, and JavaScript - edX 2024
Python Basics for Data Science - edX 2024
Critical Thinking for Software Engineers - LinkedIn Learning Community 2026

Participation in National/International Hackathons

IEEEXtreme : 17.0 2023
CodeStorm 2.0 : Yarl IT Hub 2024
UOJ Coders V4.0 : University of Jaffna 2025

Academic & Volunteering

Practical Assistant, Certificate Course in "Python for Problem Solving" 2025
Department of Computer Science, Faculty of Science, University of Jaffna

Practical Assistant, Certificate Course in "Python Essentials for Problem Solvers" 2026
Department of Computer Science, Faculty of Science, University of Jaffna

3rd Year Representative 2025
Faculty of Science, University of Jaffna

Editor 2024 – 2026
Leo Club of Nallur (Leo District 306 D12, MD 306)

Professional Membership

IEEE Student Member 2023 – 2024
IEEE Women in Engineering Member 2023 – 2024
Student Member of the Computer Society 2022 – Present
Leo Club of Nallur Member 2024 – Present

Referees

Dr. K. Thabotharan
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